

NOTOX

SPECIFIC ACTION



TOXIN
DESTROYER

Reports of In Vitro studies on Aflatoxin detoxifying ability of Notox

Trial 1

Effect of Notox addition to feed containing 375ppb Aflatoxin B1

Time interval (Hours)	Notox level, %	Analyzable Aflatoxin B ppb	Detoxification % age
24	0.025	375	0.0
	0.050	280	25.4
	0.100	95	74.7
48	0.025	328	12.7
	0.050	188	50.0
	0.100	33	91.4
72	0.025	95	74.7
	0.050	70	81.4
	0.100	0	100

* Note: Control diet moisture was 11% and aflatoxin B1 was undetectable at all the intervals. Test sample moisture varied between 25 and 27%. Experimental pH was between 6.2 and 6.5

Effect of Notox addition to feed containing 1000 ppb Aflatoxin B1

Time interval (Hours)	Notox level, %	Analyzable Aflatoxin B ppb	Detoxification % age
24	0.025	647	39.98
	0.050	518	51.94
	0.100	389	63.91
48	0.025	518	51.94
	0.050	389	63.91
	0.100	227	78.94
72	0.025	324	69.94
	0.050	216	79.96
	0.100	36	96.70

* Note : Aflatoxin B1 was undetectable in control diet at all the intervals. Sample moisture varied between 11 and 12%. Experimental pH was between 6.2 and 6.5

The above results are a mirror to the efficacy of Notox to detoxify the feed of aflatoxin B1 when mixed in the diet containing toxins, even in diets contaminated with very high levels of aflatoxin. It is advised to mix Notox at the required level depending upon the Toxin content of the ingredients in the feed and use the diet for best results.

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CONTENTS:

NOTOX contains Bio active protein moiety in fortified base.

Mycotoxins are toxic metabolites produced by filamentous fungi and moulds of three major genera of fungus like *Aspergillus*, *Fusarium* and *Penicillium*. Resultant toxic metabolites are Aflatoxins, Zearalenones (ZON), Ochratoxin A (OTA), Trichothecenes, Deoxyvalenol (DON), Fumonisins (FUM) and T2 toxins which invariably affect the chicken leading to mild to severe Hepatotoxicity, Nephrotoxicity, Neurotoxicity, production slumps, poor fertility, hatchability and immunosuppression. Uneven flock size, fatty liver conditions and poor response to vaccination and subsequent disease outbreaks are the most common problems suffered by the poultry farmers and breeders. Mould inhibitors, toxin binders and Acidifiers are not really beneficial to solve the problem efficiently. Already performed Mycotoxins and their ill effects on the vital organs like liver, and kidney's will not be contained by toxin binders.

Looking into these complications, **VESPER PHARMACEUTICALS** have introduced an unique formulation **NOTOX** with the new concept of biotransformation of the toxins and their toxic metabolites after lots of in house and field trials. **NOTOX** is a Mycotoxin destroyer that alters the molecular structure of performed **MYCOTOXINS** into non toxic metabolites. **NOTOX** also destroys **ENDOTOXINS** present in the feed.

USAGE INSTRUCTIONS:

150 gms to 250 gms per ton of feed. or as recommended by the Veterinarian.

ADVANTAGES:

Notox is specific action mycotoxin and endotoxin destroyer, elements toxic metabolites through biotransformation, is stable in G.I. tract, prevents immunosuppression, disease outbreaks in layers. **Notox** improves weight gain & FCR in broilers. **Notox** improves fertility, hatchability and number of chicks in breeders.

PRESENTATION:

25 x 1 kg in container.

NOTOX

Micotoxin contamination of feedstuffs/finished feeds is one of the serious problems faced by poultry farmers in the Country. Methods followed to reduce the severity of this problem aim at reducing the fungal growth in feedstuffs or binding the toxins or a combination of both. Though antifungal check the growth of fungi, they can not take care of preformed toxins. Hence the preferred choice is toxin binders while some farmers add antifungal also with toxin binders.

The two widely used toxin binders are aluminosilicates (Zeolite) and mannan oligosaccharide(mos). The use of aluminosilicates is quite common in poultry farms in India. Even though the toxin binding ability of aluminosilicates is high as shown in some *in vitro* trials, being a non-specific binder it may bind any one or more of the important vitamins/minerals/amino acids. Such a non-specific binding would reduce both the nutrient availability as well as the aflatoxin binding ability of zeolite. On the other hand, mannan oligosaccharide is shown to bind to the cell wall projections of bacteria and hence it's activity also may be reduced to an extent, depending upon the bacterial load.

Thus, the actual Micotoxin binding ability of mannan oligosaccharide of zeolites may be greatly reduced in the intestines, where the binding action is very essential. And apart from simple adsorption, these compounds seem to have little role in reducing the aflatoxin levels by other mechanisms. Hence, the reduction of free toxin levels to a **safe** and **harmless** levels is not assured.

Notox, developed at **Vesper Pharmaceuticals** is a product with a new concept to fight toxins. **Notox** is an aflatoxin destroyer that alters the molecular structure of aflatoxin B1 (the most harmful type of aflatoxin for poultry) in to non-toxic metabolites like aflatoxicol and aflatoxin B2b and then binds on to the altered molecules. A stabilized protein moiety, **Notox** is specific in action, stable under the gastrointestinal conditions. **Notox** does not react with any other chemicals/molecules. **Notox** doubly assures the safety of your birds.



Differentiates the Difference

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